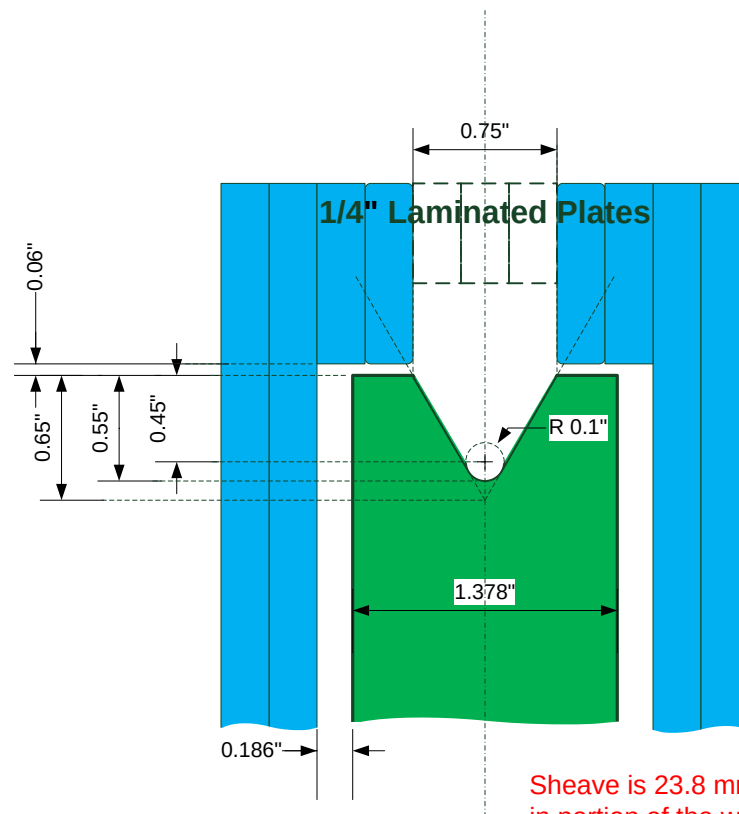
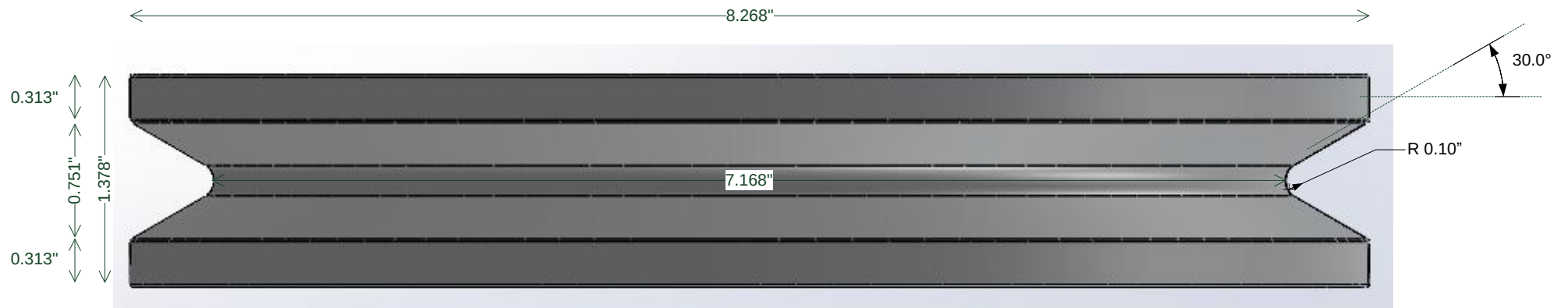
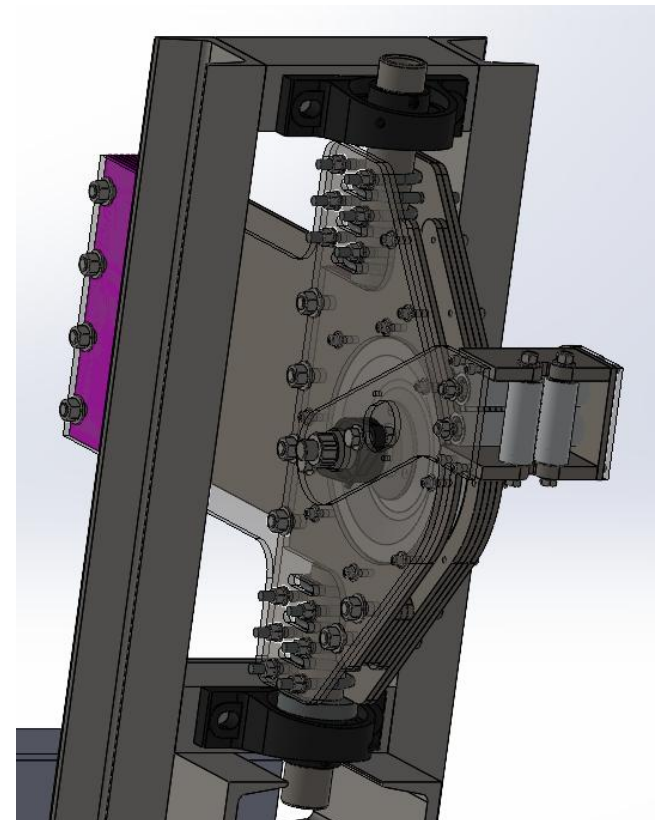
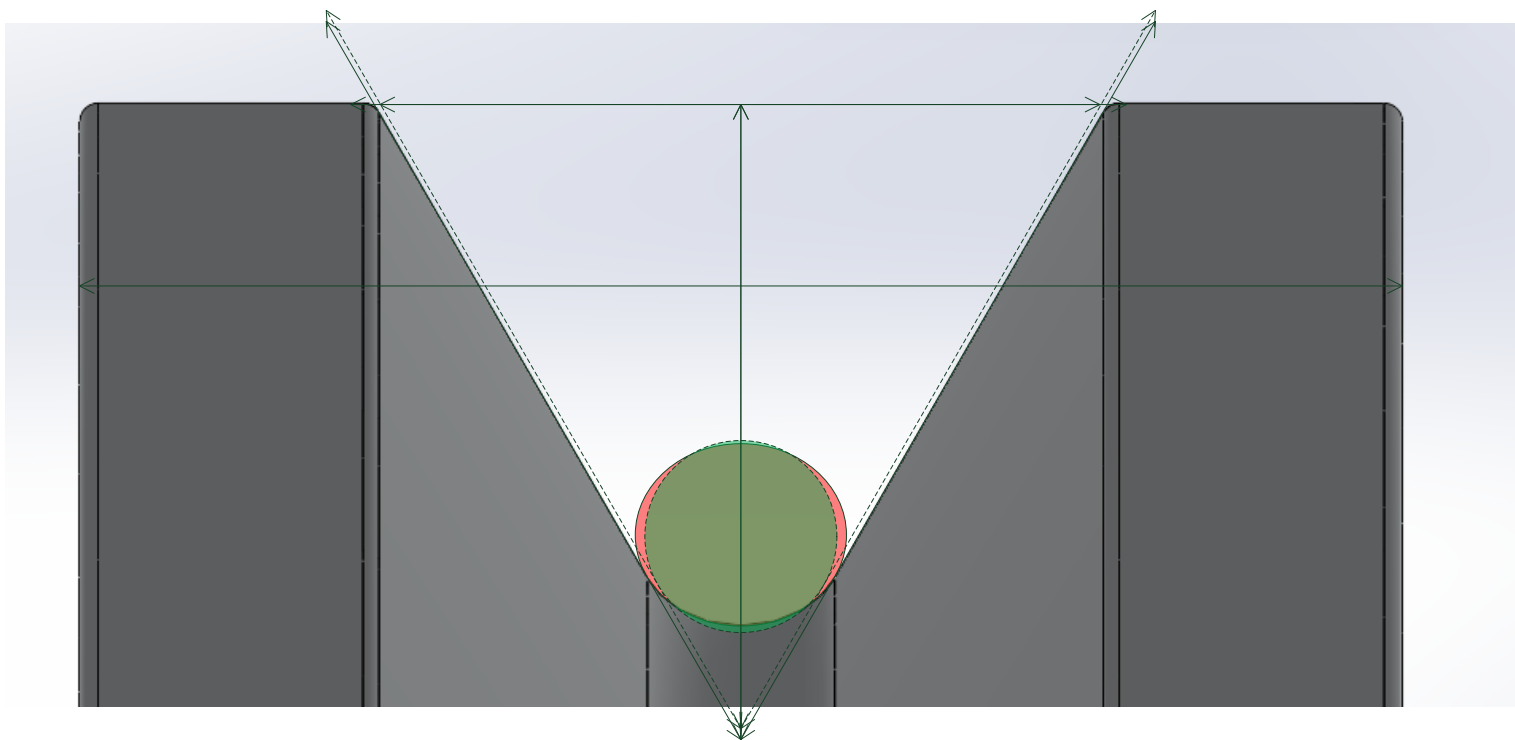
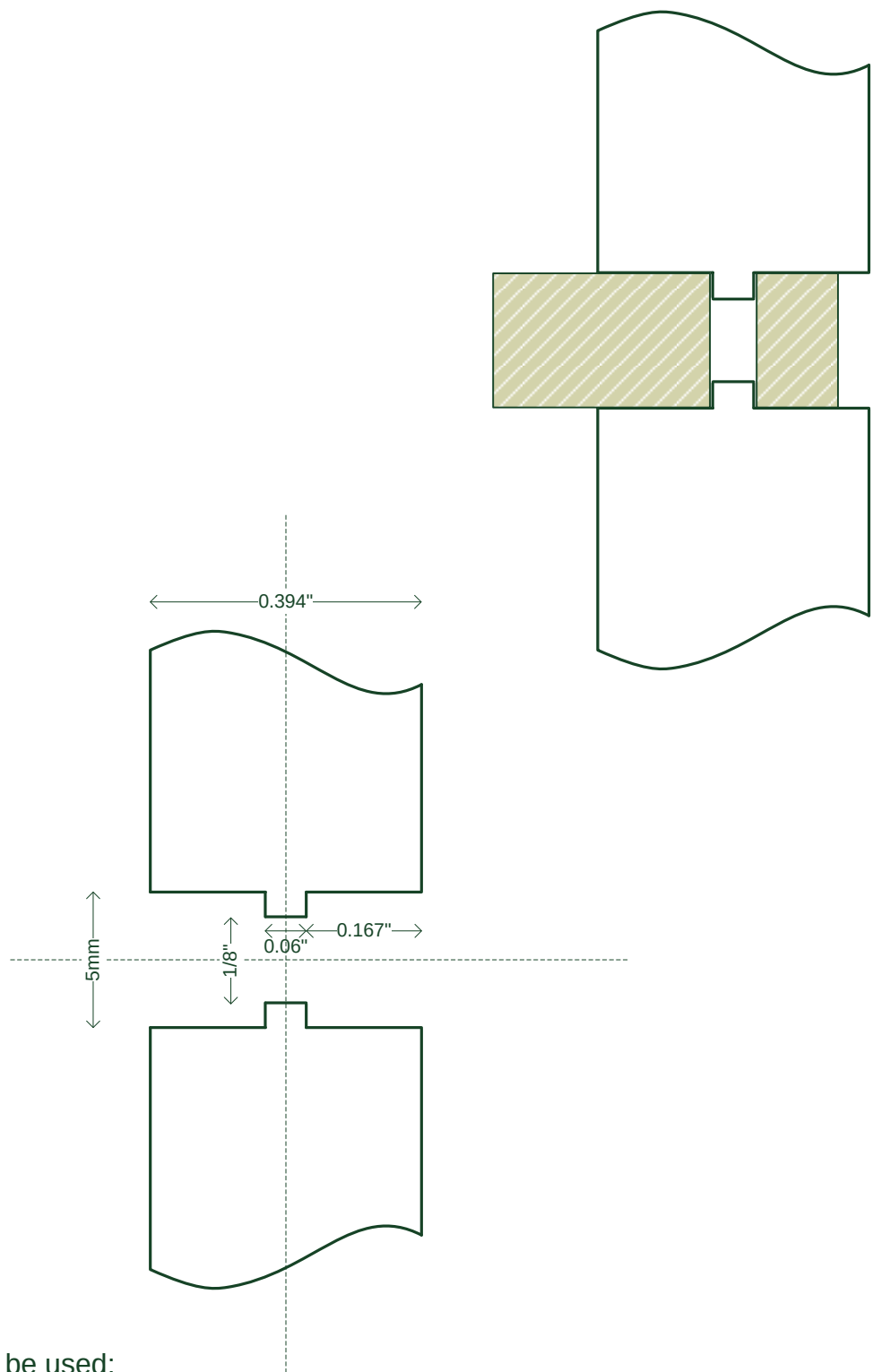




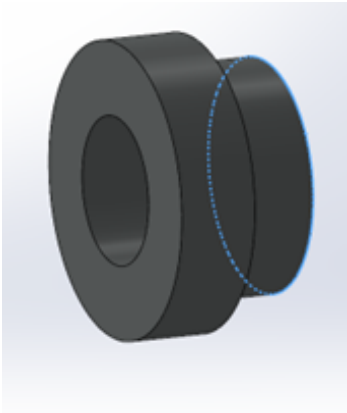
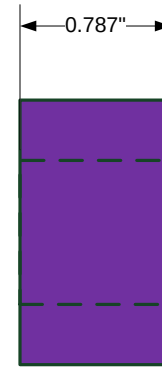
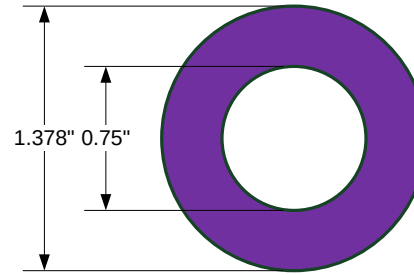
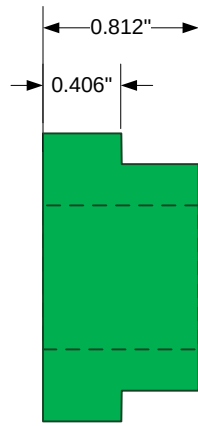
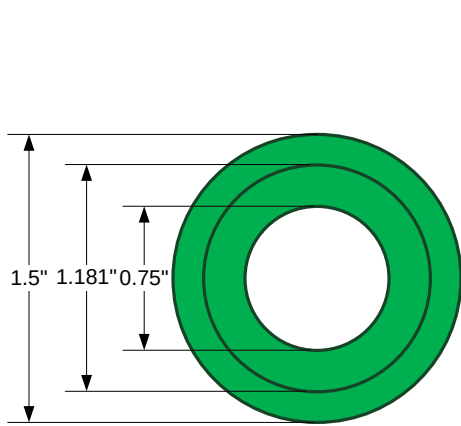
Sheave is 23.8 mm thick in portion of the web that is nearest its bore ID. That is the thickness of its press-fit bearing





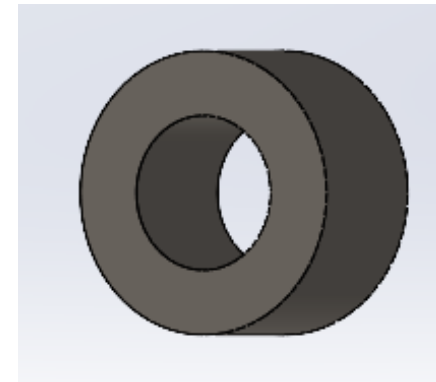


Magnetic Sensor to be used;  
Honeywell SR3-A1  
Available from Mouser and  
others



Starting Stock: 1.5x0.75 DOM Steel

2 Each Required



Need two more 35 mm (3/8") bushing to go on outside of fairlead shaft. 20 mm tall.

Looks like Online Metals has 1.375x0.75 DOM tubing. (Have to look for 0.25 Wall.) Length not critical so maybe cut myself.

Honeywell  
Hall Sensor

1/4" Laminated Plates

Headed Dual  
HE Sensor

4.87"

UCFL204-28  
Flange Bearing

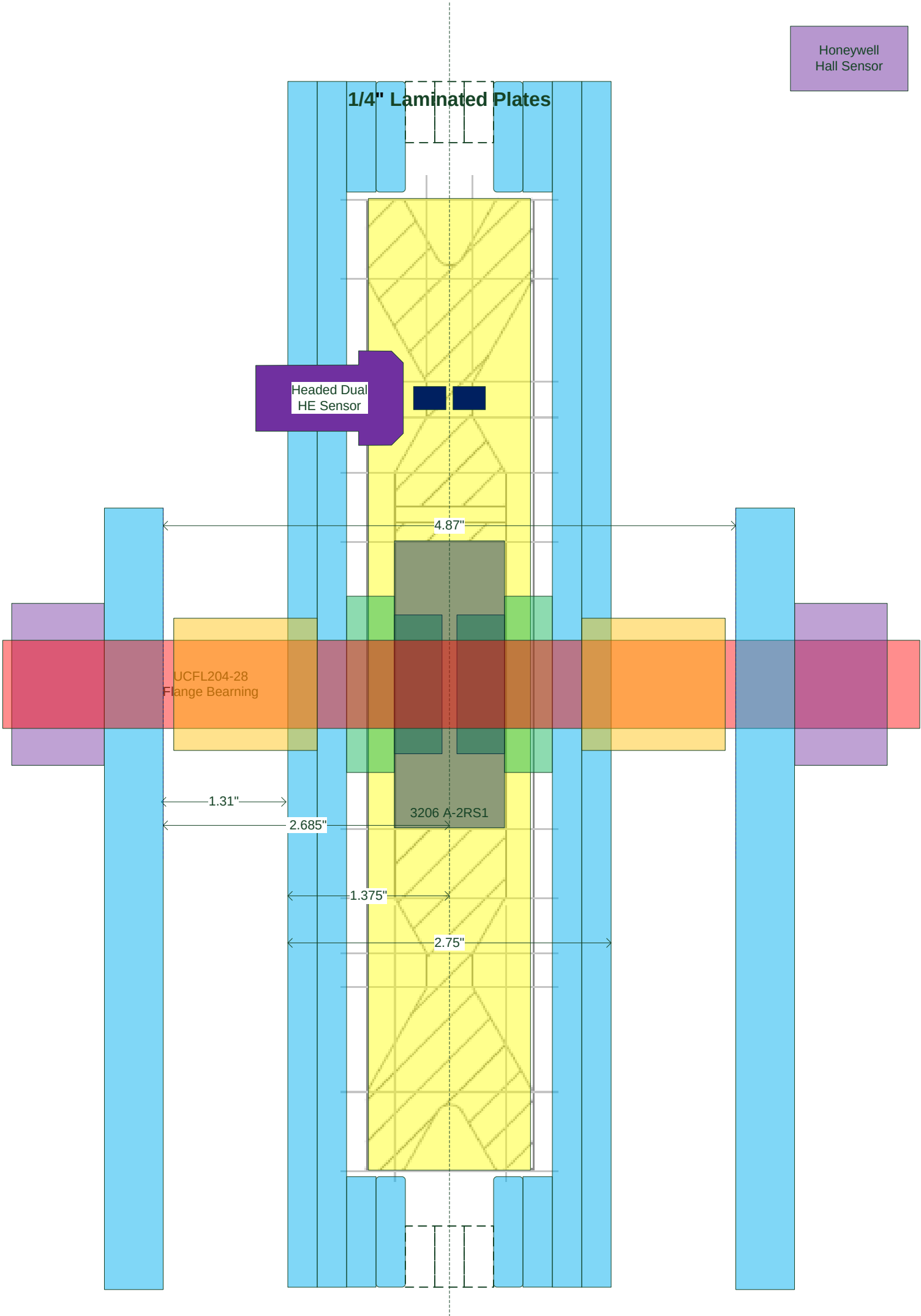
3206 A-2RS1

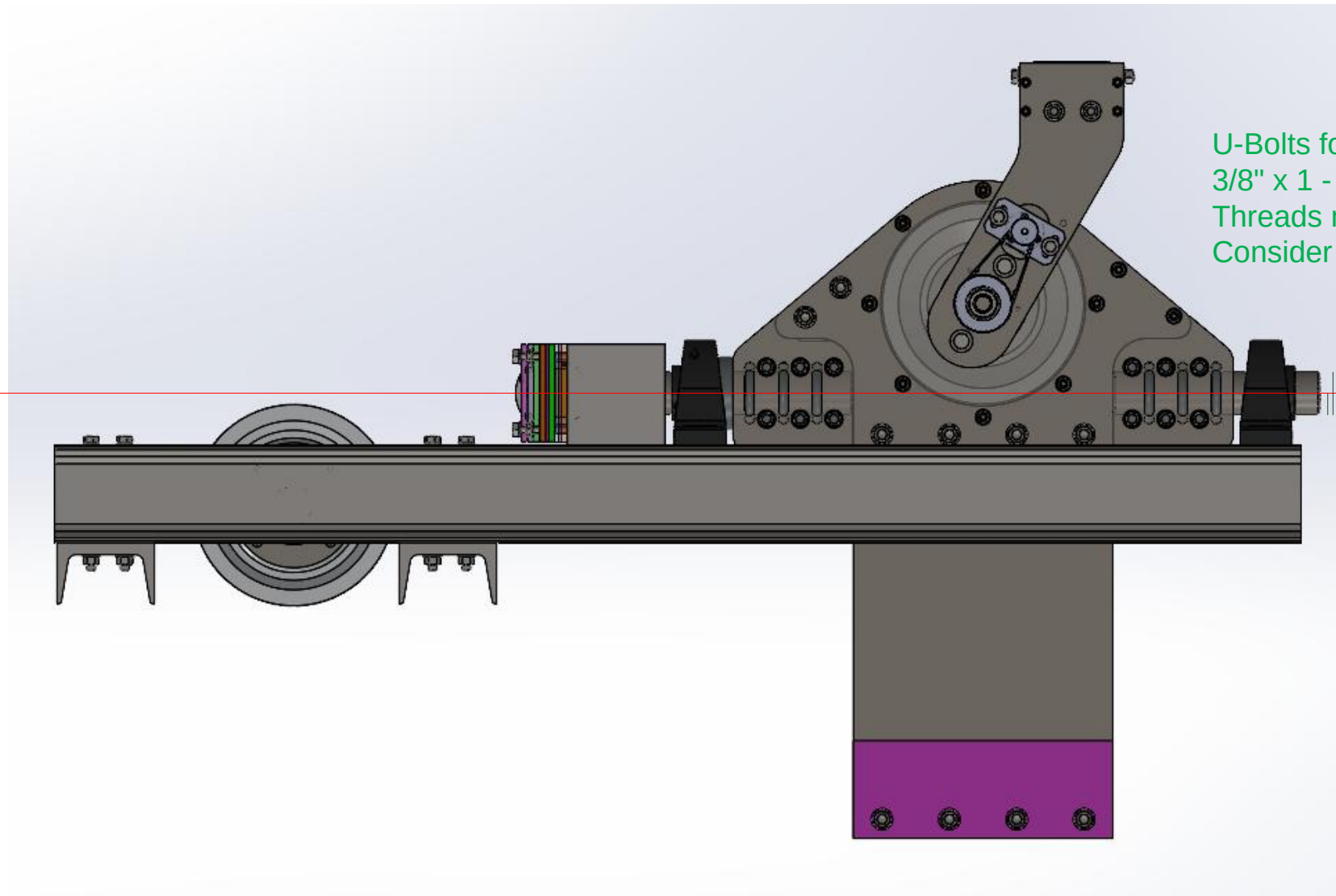
1.31"

2.685"

1.375"

2.75"





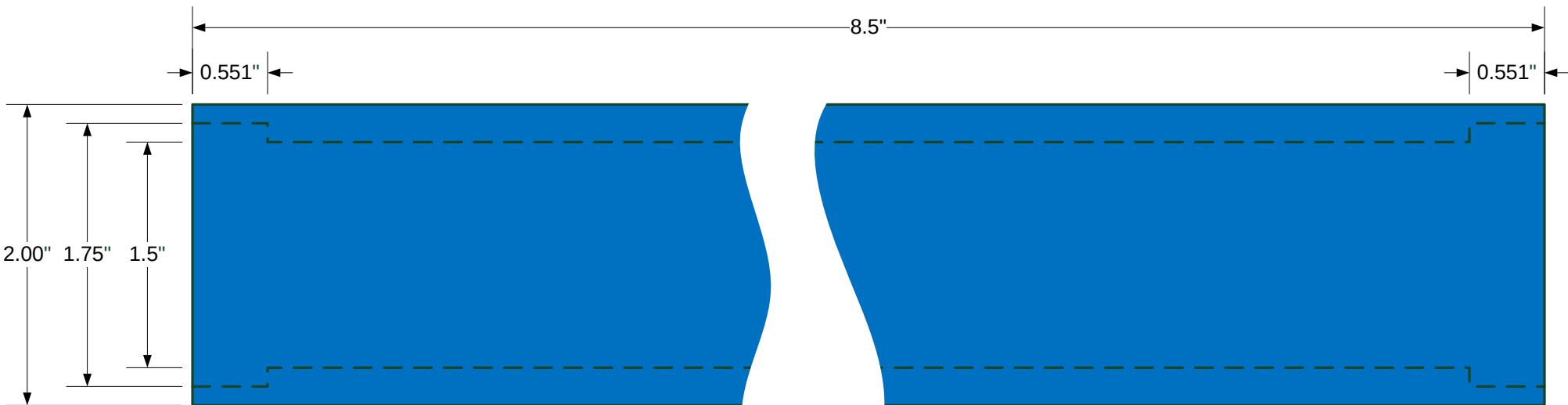
U-Bolts for Clamping DOM Pipe  
3/8" x 1 - 3/4" x 2.5"  
Threads must begin by 2.25"  
Consider coating

Note the Tension Sheave is not properly positioned for  
the rope to traverse the iFGT's Elements!!!!



Yellow denotes area to be threaded for  $\frac{3}{4}$ -10 nut.

Note: Fairlead sheave is not centered on this shaft

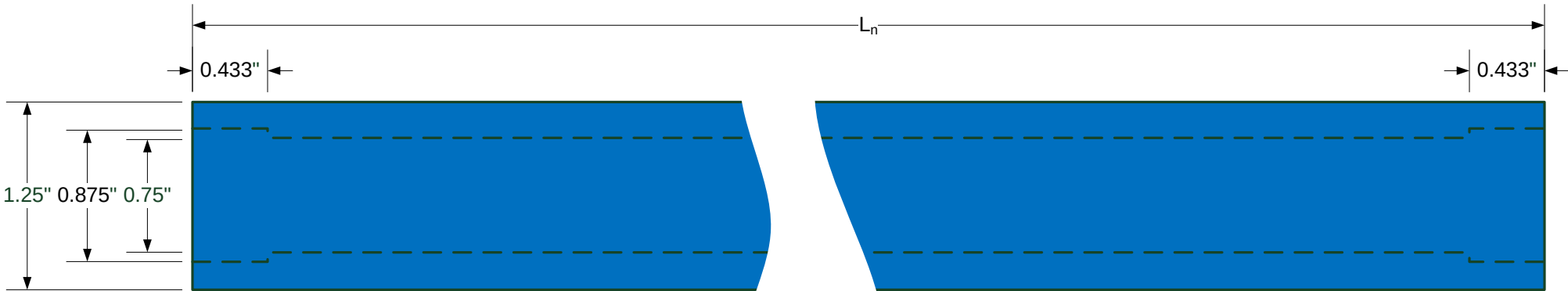
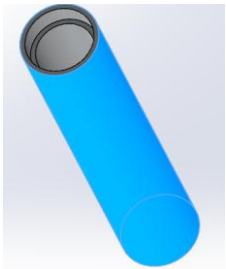


Bearings: 1635 2RS-1  
 OD     1.75"  
 ID     0.75"  
 Width  0.5"

2 Rollers Required

DOM tubing for bushings: Best bet seems to be 1" OD x 0.760" ID

6061-T6 Aluminum Extruded Tubing

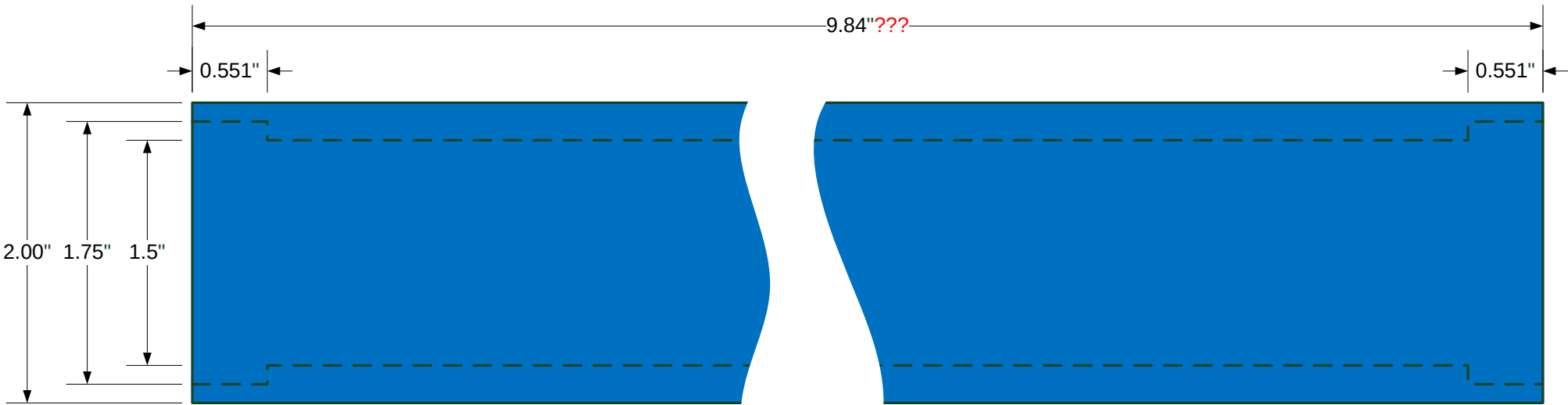


Bearings: D/W R6-2R1  
 OD     0.875"  
 ID     0.375"  
 Width  0.2812"

DOM tubing for bushings: Best bet seems to be 0.5" OD x 0.370" ID

| Roller                    | Length | Number |
|---------------------------|--------|--------|
| L <sub>1</sub> (Fairlead) | 3.150" | 2      |
| L <sub>2</sub> (Fairlead) | 4.625" | 2      |
| L <sub>3</sub> (Xwind)    | 5.250" | 2      |

Revisit: Short Roller Lengths



Bearings: 1635 2RS-1  
 OD 1.75"  
 ID 0.75"  
 Width 0.5"

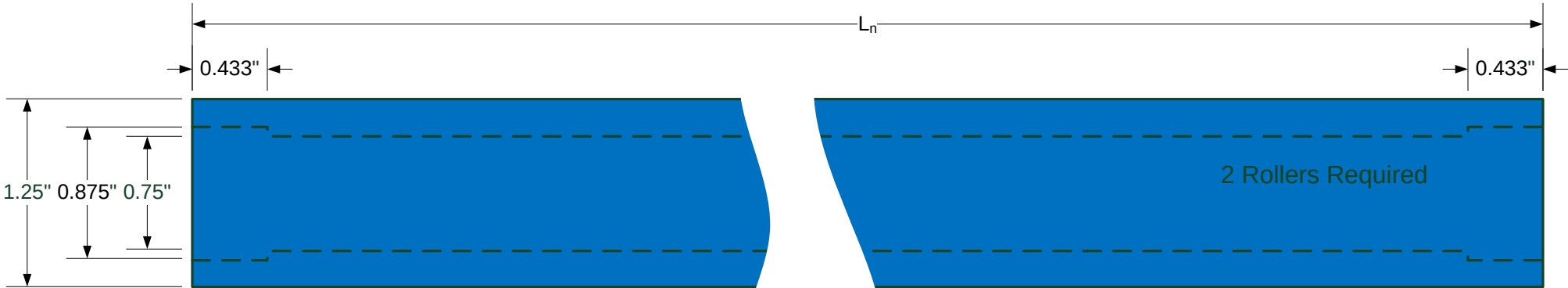
Revisit: Reduce Length?

2 Rollers Required

Need DOM steel derived  
 bushing spacers for all rollers.

DO NOT USE  
 Original Mario dimensions.  
 Use Revised instead.

6061-T6 Aluminum  
 Extruded Tubing

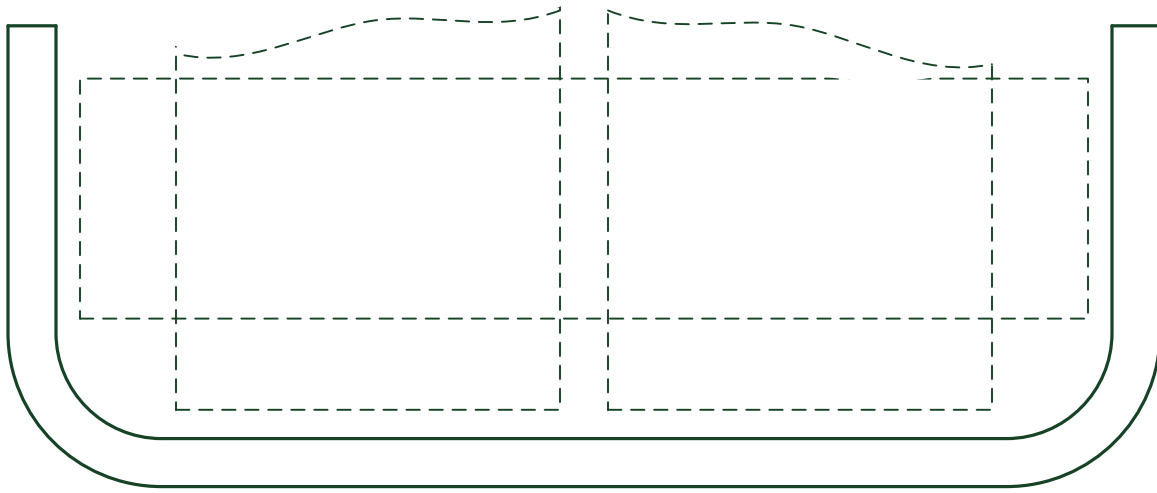


Bearings: D/W R6-2Z  
 OD 0.875"  
 ID 0.375"  
 Width 0.2812"

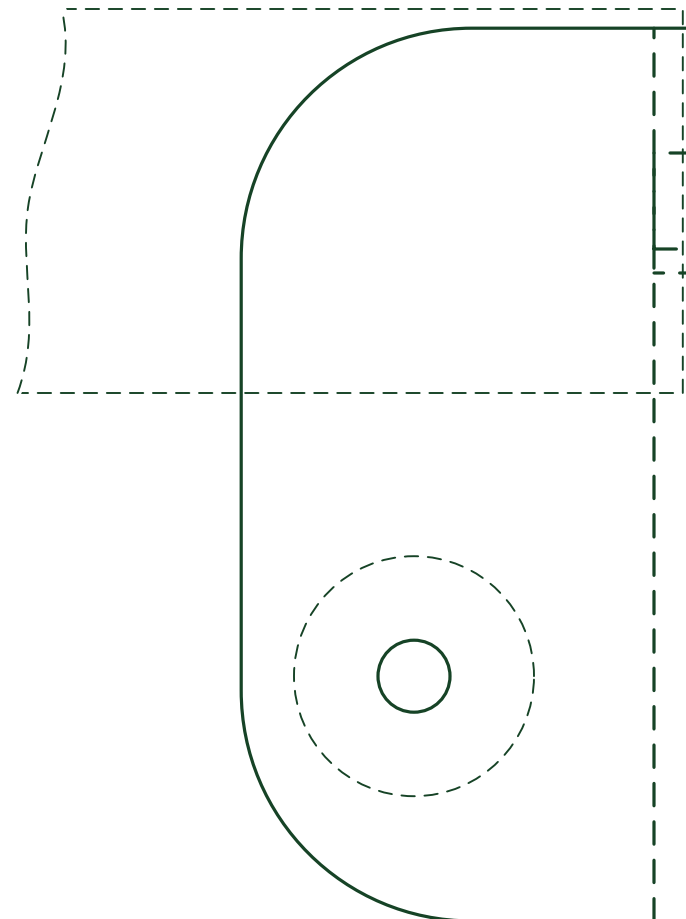
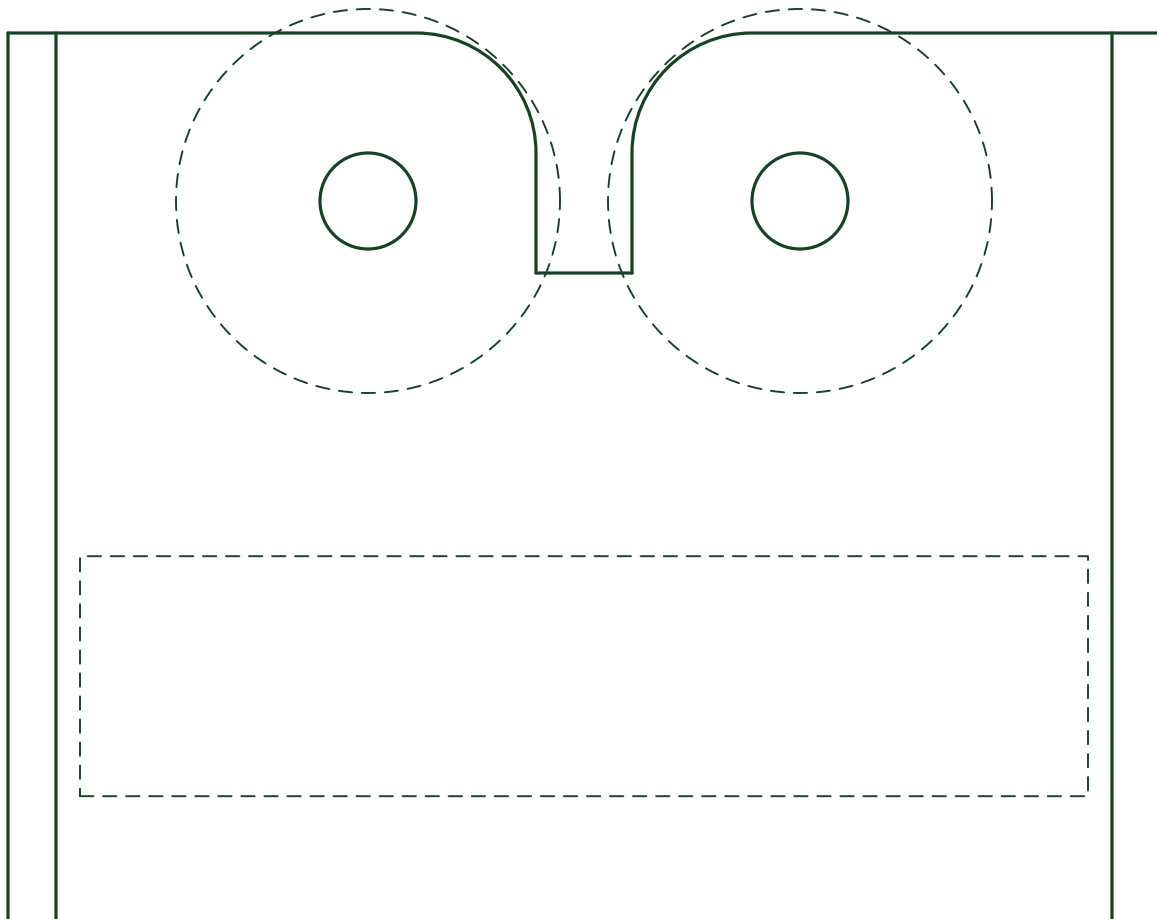
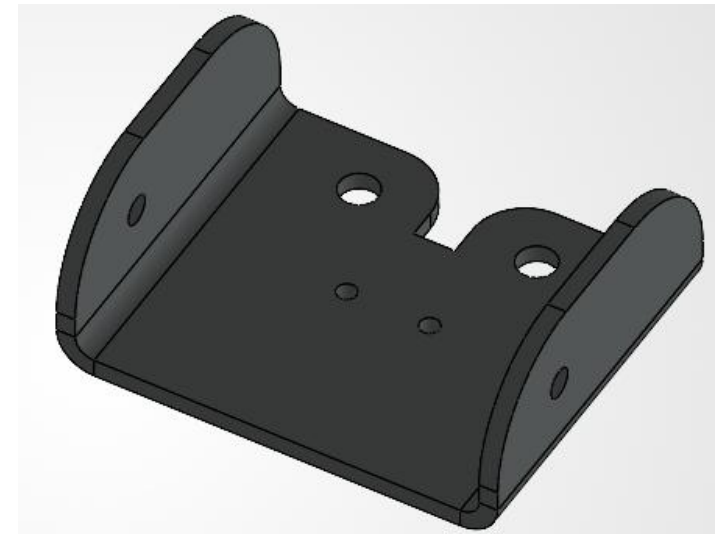
Revisit: Review bushing  
 lengths.

| Roller | Length | Number |
|--------|--------|--------|
| $L_1$  | 3.150" | 2      |
| $L_2$  | 5.213" | 4      |

2 Rollers Required



Needs placement  
finalized and dimensions



| Item       | Size      | Number | Notes                                          |
|------------|-----------|--------|------------------------------------------------|
| Cap Screws | 0.25x1.25 | 20     |                                                |
| Cap Screws | 0.25x1.00 | 8      | May likely be 1.25" long to combine with above |

To Do:

Fairlead

Will need to cut holes in the outer fairlead plates centered on the axle to allow the UCFL204-12 pillow block bearings that support its crossed roller arms to somewhere somewhere on the order of 2" to allow the pillow block's bearing to rest on the next plate inside. This was an oversight by Mario. The actual bearing OD is 1.25" but the set screws will need clearance to fit and be tightened.

Need ~1" hole through plates for dual magnetic sensor plug

Need to tap for the 0.25x1.25 inch cap screws that tie together all the laminated plates.

Also will need to drill and tap for the plates that hold the crossed rollers